

## Day 23 Topic

# Chain-of-Thought (CoT) Prompting + Reasoning in LLMs

## Introduction

In 2026, AI is no longer just:

- text generation
- chatbot conversations
- content writing

Modern frontier models like:

- OpenAI GPT-5 series
- Anthropic Claude Sonnet reasoning models
- Google Gemini reasoning models
- DeepSeek DeepSeek reasoning systems

are becoming:

## reasoning systems

These models:

- plan
- analyze
- self-correct
- use tools
- break problems into steps
- solve multi-stage workflows

This shift happened because of:

## Chain-of-Thought Reasoning

# Chain-of-Thought Reasoning

## ◆ 1. What is Reasoning in LLMs?

### ✗ Biggest Misconception

People think:

Plain text

LLMs think like humans.

This is incorrect.

### ✓ Correct Understanding

LLMs are:

#### 👉 probabilistic token prediction systems

They predict:

- next words
- next reasoning steps
- likely logical patterns

### 📌 Definition

Reasoning in LLMs means:

Generating intermediate logical steps to solve complex problems.

## Example

Question:

Plain text

A shop sold 25 books on Monday and 30 on Tuesday.  
How many books were sold in total?

To solve this, model must:

1. understand question
2. extract numbers
3. apply arithmetic logic
4. generate final answer

This process is:

👉 reasoning

## ◆ 2. Why Reasoning Became Important After 2024

Earlier LLMs:

- generated fluent text
- but often failed at:
  - math
  - planning
  - logic
  - coding workflows

### Modern 2025–2026 Models

Modern reasoning models:

- think longer
- use internal reasoning traces
- perform self-correction
- plan before responding

Recent GPT-5 reasoning systems can:

- allocate extra “thinking time”
- generate internal planning stages
- improve tool usage & multi-step workflow

## ◆ 3. What is Chain-of-Thought (CoT) Prompting?

### ✓ Definition

Chain-of-Thought Prompting means:

Encouraging the model to generate intermediate reasoning steps before the final answer.

### Core Idea

Instead of:

Plain text

Give answer directly.

We guide model:

Plain text

Let's think step by step.

## ◆ 4. Why CoT Became Revolutionary

One of the biggest discoveries in modern AI:

A simple phrase:

Plain text

Let's think step by step.

dramatically improved:

- reasoning
- arithmetic
- logic
- planning



## ◆ 5. Standard Prompting vs CoT Prompting

### ✗ Standard Prompting

Plain text

If 12 cars each have 4 tires,  
how many tires are there?

Weak model may guess:

Plain text

36

Wrong.

### ✓ Chain-of-Thought Prompting

Plain text

If 12 cars each have 4 tires,  
let's think step by step.

Output:

Plain text

Each car has 4 tires.  
 $12 \times 4 = 48$ .  
So total tires = 48.

Correct.

## ◆ 6. Why CoT Works Internally

LLMs generate text sequentially.

CoT forces model to:

- slow down reasoning
- generate intermediate steps
- stabilize token prediction
- reduce logical jumps

### 🧠 Important Insight

CoT does NOT magically create intelligence.

Instead:

👉 it improves reasoning trajectory.

## ◆ 7. Real-Life Analogy

Imagine a teacher asking:

### ✗ Direct Question

Plain text

Give answer immediately.

Student may guess.

### ✓ Better Instruction

Plain text

Show all reasoning steps.



## ◆ 8. Types of Reasoning in Modern LLMs

| Reasoning Type | Example                    |
|----------------|----------------------------|
| Mathematical   | Calculations               |
| Logical        | Puzzle solving             |
| Commonsense    | Real-world understanding   |
| Multi-Hop      | Connecting multiple facts  |
| Planning       | AI agents & workflows      |
| Tool Reasoning | Deciding which tool to use |

## ◆ 9. Mathematical Reasoning

### 🔍 Example

Plain text

A train moves 60 km/h for 5 hours.  
How far does it travel?

Reasoning:

Plain text

Distance = Speed × Time  
 $60 \times 5 = 300$  km

## ◆ 10. Commonsense Reasoning

### 🔍 Example

Plain text

Can an elephant fit inside a refrigerator?

Model uses:

- physical understanding
- size relationships
- real-world knowledge

## ◆ 11. Multi-Hop Reasoning

### 🔍 Example

Plain text

Who is the CEO of the company that owns Instagram?

Model must:

1. know Instagram owned by Meta
2. know Meta CEO

This is:

👉 multi-hop reasoning

## ◆ 12. Zero-Shot CoT

### ✓ Definition

No reasoning examples provided.

Only instruction:

Plain text

Let's think step by step.

### 🔍 Example

Plain text

John has 10 apples.  
He gives away 4.  
How many remain?  
Let's think step by step.

## ◆ 13. Few-Shot CoT

### ✓ Definition

Provide reasoning examples before actual problem.

### 🔍 Example

Plain text

Q:  $2 + 3 = ?$   
A: First add 2 and 3.  
Answer = 5.

Q:  $5 + 7 = ?$   
A:



## 🧠 Why It Helps

Model learns:

- reasoning structure
- response style
- logical sequence

## ◆ 14. Emergent Abilities (VERY IMPORTANT)

### ✓ Definition

Capabilities that suddenly appear when models become sufficiently large.

## 🔥 Key Discovery

Reasoning ability does NOT grow linearly.

Sometimes:

- arithmetic
- planning
- coding
- logic

appear suddenly at scale.

## 📌 Example

Small models:

### ◆ 15. Why Larger Models Reason Better

Larger models:

- store more patterns
- learn more logical relationships
- develop richer representations

### 📌 Result

Large models:

- decompose tasks better
- self-correct better
- maintain longer reasoning chains

GPT-5 reasoning models introduced extended reasoning & multi-step tool planning capabilities. OpenAI +2

### ◆ 16. Modern 2026 Reasoning Models

Modern AI systems are shifting toward:

#### 👉 dedicated reasoning models

Examples:

- GPT-5 reasoning variants
- Claude reasoning models
- DeepSeek-R1 systems

These models:

- think longer
- plan internally
- perform deeper analysis
- execute tool workflows

### ◆ 17. Hidden Reasoning / Internal CoT

IMPORTANT 2026 TREND.

Modern frontier models often:

- hide internal reasoning traces
- use private scratchpads
- optimize reasoning internally

Why?

- safety
- efficiency
- jailbreak prevention
- reasoning reliability

### ◆ 18. Self-Consistency Prompting

Advanced reasoning method.

#### ✅ Idea

Generate:

- multiple reasoning paths

Select:

- most consistent answer.

#### 🔍 Example

If outputs are:

Plain text

42  
42  
42  
36  
42



## ◆ 19. Tree of Thoughts (ToT)

Advanced reasoning architecture.

### ✅ Definition

Instead of:

- one reasoning chain

Explore:

- multiple reasoning branches.

### 🧠 Human Analogy

Humans often:

- explore multiple ideas
- compare solutions
- evaluate options

ToT imitates this process.

## ◆ 20. CoT in AI Agents

Modern AI agents require:

- reasoning
- planning
- memory
- tool usage

Without reasoning:

agents fail at complex workflows.

## ◆ 21. ReAct Prompting (Bridge to Next Topic)

ReAct =

### 👉 Reason + Action

Model:

1. reasons
2. performs action
3. observes result
4. reasons again

This powers:

- autonomous agents
- coding agents
- research systems

## ◆ 22. Hallucination in Reasoning

VERY IMPORTANT.

Even reasoning models can:

- produce incorrect logic
- sound extremely confident
- hallucinate reasoning chains

### 📌 Important Understanding

❌ Detailed reasoning ≠ correctness

Modern GPT-5.5 systems reduced hallucinations significantly compared to earlier models.



## ◆ 23. Cost & Latency of Reasoning

Reasoning models:

- generate more tokens
- think longer
- consume more compute

### Tradeoff

| Benefit          | Cost            |
|------------------|-----------------|
| Better reasoning | Higher latency  |
| Better planning  | More tokens     |
| Better accuracy  | Higher API cost |

Recent research shows incomplete reasoning chains can significantly hurt performance in reasoning-specialized systems. 

## ◆ 24. Real-World Applications

### Coding Agents

- debugging
- multi-file reasoning
- code planning

### AI Tutors

- step-by-step explanations
- math solving